

Maciej Czerniak

Data Science & AI Student

Machine Learning, Data Analytics & Applied AI

Breda, Netherlands

+31 615580222

✉ maciek96czerniak@gmail.com

🐙 GitHub

🌐 LinkedIn

📁 Portfolio

PROFILE

Data Science and Artificial Intelligence student at Breda University of Applied Sciences. Building applied AI and software systems across computer vision, geospatial analysis, robotics, LLM-backed query assistants, and Python package development. Currently working as a Student Assistant Developer on Tangible Landscape systems combining depth sensing, GRASS GIS, real-time sensor processing, and interactive visualization.

EXPERIENCE

•Breda University of Applied Sciences

Nov. 2024 – Present

Student Assistant Developer

Breda, Netherlands

- Developing Python-based Tangible Landscape systems connecting physical terrain models with real-time geospatial analysis, depth sensing, GIS tooling, and visualization.
- Built and maintained Python workflows for real-time sensor processing, geospatial analysis, and interactive visualization.
- Integrated and adapted depth-sensor pipelines, including the transition from Xbox Kinect to Orbbec Femto Bolt.
- Worked on low-level C++ changes for the GRASS GIS `r.in.kinect` module where hardware integration required it.
- Built interactive geospatial features including tree-planting, viewshed analysis, and topography simulations.

EDUCATION

•Breda University of Applied Sciences

Sep. 2024 – Present

BSc Applied Data Science and Artificial Intelligence

Breda, Netherlands

- Relevant areas: machine learning, data analytics, statistics, computer vision, robotics, software engineering, databases, and applied AI projects.

•Zespół Szkół Budowlanych im. Eugeniusza Kwiatkowskiego

2012 – 2016

High School Diploma: Building Technician

Konin, Poland

PROJECTS

•NPEC Root Inoculation Pipeline

2025 – 2026

Computer vision and robotics pipeline for plant root analysis and robotic inoculation

- Developed multiclass image segmentation workflows for root, shoot, seed, and background classes.
- Built root system architecture extraction logic to identify plant instances, primary roots, and biologically relevant target points.
- Connected image-based analysis with simulated robotic control for an Opentrons OT-2 liquid-handling robot.
- Used Python, TensorFlow/Keras, OpenCV, image processing, evaluation metrics, PyBullet, and experiment tracking concepts.

•Text-to-SQL Query Assistant

2025 – 2026

LLM-backed assistant for converting natural language questions into SQL queries

- Built a prototype assistant that converts business questions into SQL queries using locally hosted LLMs.
- Connected the system to relational databases and tested generated queries against structured business data.
- Improved prompting and validation rules to reduce invalid SQL and make outputs more reliable.
- Used Python, SQLAlchemy, Pandas, MySQL/PostgreSQL/SQLite, Ollama, Streamlit, and prompt engineering.

•Sign Language Recognition Package

2026

Python package refactor for model training, evaluation, testing, and deployment preparation

- Refactored notebook-based training code into reusable Python package modules.
- Added configuration, data loading, preprocessing, model training, evaluation, logging, and output persistence.
- Improved package quality with unit tests, documentation structure, and development workflow changes.
- Used Python, PyTorch, Poetry, pytest, Sphinx, Git, GitHub, and Weights & Biases concepts.

TECHNICAL SKILLS

Languages: Python, SQL

AI & Data: machine learning, deep learning, computer vision, image segmentation, model evaluation, data preprocessing

Libraries: Pandas, NumPy, Matplotlib, OpenCV, TensorFlow/Keras, PyTorch, scikit-learn, SQLAlchemy

Systems & Tools: Git, GitHub, Poetry, pytest, Sphinx, Docker, Streamlit, FastAPI basics, W&B

Geospatial & Sensors: GRASS GIS, Tangible Landscape, interactive visualization

Databases: MySQL, PostgreSQL, SQLite